

Auteur: Stan Van Campenhout
Studierichting: Informatica
Oefeningenreeks 4A nr. 53.5

$$\begin{aligned}\int \sin 3x \cos^{-4} 3x dx &= \frac{1}{3} \int \sin 3x \cos^{-4} 3x d(3x) \\&= -\frac{1}{3} \int \cos^{-4} 3x d(\cos 3x) \stackrel{\cos 3x=t}{=} -\frac{1}{3} \int \frac{1}{t^4} dt \\&= \frac{1}{9t^3} + c \stackrel{t=\cos 3x}{=} \frac{1}{9 \cos^3 3x} + c\end{aligned}$$

References